Higher National Diploma in Quantity Surveying
Second Year, First Semester Examination – 2018
QS 21054 – Civil Engineering Construction ii for QS

Instructions for Candidates:	No. of questions : 06
Answer any **Five (05)** questions.	No. of pages : 05
All question carry equal marks	Time : Three (03) hours

1. You are newly appointed as a QS in the ABC Company. You have been asked to submit a report about the quality of the cement which is brought to the site.
   a. State the raw materials used to manufacture of cement. (02 marks)
   b. State four important properties of cement. (04 marks)
   c. What type of cement do you propose for construction in costal belt area and justify Your answer. (04 marks)
   d. In that report, explain the steps you have taken to check the quality of the cement. (10 marks)
2. Concrete is the universal material of the construction.
   a. State the four tests that you would recommend to check the workability and strength of concrete. (04 marks)
   b. Explain why compaction of concrete is important and how you perform the compaction. (05 marks)
   c. Explain why curing is important and how you would cure the concrete work. (05 marks)
   d. What do you understand by the pre-stressed concrete? and list down the two advantages and disadvantages of pre-stressed concrete. (06 marks)

3. Most of the heavy construction sites are using the Ready mix concrete. You are appointed as the in charge of the site.
   a. State five important consideration when the methods of placement of ready Mixed Concrete. (05 marks)
   b. What are the advantages and disadvantages of Ready Mixed Concrete? (05 marks)
   c. Plastic is a very popular as a construction material in the field of civil engineering. Use of plastic formwork for concrete construction has many advantages such as

durability, cost, and flexibility compared to the other material such as timber, steel and aluminium. Justify this statement. (05 marks)

d. You are appointed as a QS in the ABC site and you are asked to give the necessary instructions to a trainee regarding the removal time of formwork of the concrete. Ordinary Portland cement is used for this purpose. Instruct the trainee with the necessary explanation in a form of a chart. (05 marks)

4. Most of the mechanical components of the building services have a relatively shorter life span than the building structure.
   a. Justify this statement considering the water supply, electricity and air condition. (05 marks)
   b. Explain the term "sick building syndrome" with poor ventilation. (05 marks)
   c. State the ventilation systems and explain those. (05 marks)
   d. If you are asked to plan a water service in a wash room in a boys' hostel, identify the necessary accessories need to complete this task. (05 marks)

5. Concrete mix design is economically proportioning of ingredients for better strength and durability of concrete
   a. State the main importance of the mix design. (05 marks)
   b. State the factors influencing the choice of mix design. (05 marks)
   c. The given specification of the concrete which are target mean strength of 43 N/mm^2 for 28 days, the type of cement is ordinary Portland cement, the structure is moderated exposure condition with aggregate 20mm coarse crushed and fine unerushed aggregate and workability is measured with slump test is 25 mm, the bulk specific gravity of 2.7, 60% of fine aggregate passes the 600 μm sieve. You are asked to find out necessary quantity of ingredients to produce 100m3 of concrete. (Necessary charts are attached). (10 marks)

6.
   a. Explain the differences between rigid frame and Braced frame with examples. (05 marks)
   b. State the type of frame structures which are introduced in the construction industry. (05 marks)
   c. State the difference between load bearing structures and framed structures. (05 marks)
   d. Transmission towers are constructed with steel frame. Explain this statement. (05 marks)